

Aptitude, 2017 EC, entrance exam, version 2

Grade 12 · Aptitude · 2017 E.C.

Time allowed: 120 minutes

Questions: 55

Total marks: 55

Name _____ Grade _____ ID No. _____ Date _____

1. CALF : COW :: LAMB : ? (1)

A. Sheep	B. Deer
C. Horse	D. Horse
2. BOOK : PUBLISHER :: FILM : ? (1)

A. Writer	B. Editor
C. Director	D. Producer
3. EXTROVERTED (1)

A. Shy	B. Outgoing
C. Deceiver	D. Energetic
4. ASSERTIVE (1)

A. Passive	B. Aggressive
C. Confident	D. Hesitant
5. DISINTERESTED (1)

A. Impartial	B. Biased
C. Bored	D. Unconcerned
6. SUBJUGATE (1)

A. Liberate	B. Oppress
C. Protest	D. Conquer
7. PENNYWISE (1)

A. Selfish	B. Pleasant
C. Ungrateful	D. Extravagant
8. FRIENDLY (1)

A. Hostile	B. Warm
C. Supportive	D. Affectionate
9. Which word is different? (Apple, Carrot, Grape, Banana) (1)

A. Apple	B. Carrot
C. Grape	D. Banana
10. Which word is different? (London, Paris, New York, California) (1)

A. London	B. Paris
C. New York	D. California

-
11. Which word is different? (Lion, Tiger, Elephant, Leopard) (1)
- A. Lion B. Tiger
C. Elephant D. Leopard
-
12. The engineer's meticulous attention to detail in the design of the building ensured that everything was perfectly executed, from the architecture to the finishing touches. (Underlined: meticulous) (1)
- A. reckless B. thorough
C. curious D. imprecise
-
13. As soon as all the indispensable forms have been filled out, we will process our applications. (Underlined: indispensable) (1)
- A. requisite B. nonessential
C. tentative D. imprecise
-
14. Ali is all the time gluttonous to explore a wide range of topics, seeking to learn more and never satisfied with just the basics. (Underlined: gluttonous) (1)
- A. voracious B. excessive
C. limited D. disinterested
-
15. I've always been somewhat introverted, preferring to reflect quietly rather than engage in constant chatter with friends. (Underlined: introverted) (1)
- A. outgoing B. reserved
C. indifferent D. verbose
-
16. She wanted to impress all her dinner guests with the food she served, so she carefully studied the necessary culinary arts. (Underlined: culinary) (1)
- A. architectural B. literary
C. cooking D. gardening
-
17. Correct sentence: Neither the players nor the manager be ready for the next press conference. (1)
- A. Neither the players nor the manager be ready for the next press conference.
B. Neither the players nor the manager were ready for the next press conference.
C. Neither the players nor the manager is ready for the next press conference.
D. Neither the players nor the manager have been ready for the next press conference.
-
18. Correct sentence: I personally believe the team has played wonderful in the match. (1)
- A. I personally believe the team has played wonderful in the match.
B. I personally believe the team have played wonderful in the match.
C. I personally believe the team has played wonderfully in the match.
D. I personally believe the team played wonderful in the match.
-
19. Correct sentence: Ali is one of the authorities who is responsible for the decision. (1)
- A. Ali is one of the authorities who is responsible for the decision.
B. Ali is one of the authorities who are responsible for the decision.
C. Ali is one of the authority who is responsible for the decision.
D. Ali is one of the authority who are responsible for the decision.

-
20. *Keneni has to study four books (English, Physics, Chemistry and Biology) on four different days from Thursday to Sunday. The order of the books is as follows: Chemistry is studied on day before the day of English. Biology is studied on day after the day of Physics.* (1)

Keneni has to study four books (English, Physics, Chemistry and Biology) on four different days from Thursday to Sunday. The order is: Chemistry is studied on day before the day of English. Biology is studied on day after the day of Physics. Which of the following is a correct order of books?

- A. Chemistry, Biology, English, Physics B. Biology, Chemistry, English, Physics
C. Physics, Chemistry, Biology, English D. English, Chemistry, Biology, Physics
-

21. *Same conditions.* (1)

Which one of the following is NOT possible?

- A. Chemistry on Thursday B. Physics on Sunday
C. Physics on Thursday D. Biology on Saturday
-

22. After hearing about the new job opening in his field, Mark decided to apply immediately. Though it would mean moving to a new city, he was excited about the opportunity and felt it was a great career move. The job was also offering a higher salary than his current position, which made the decision easier for him. What is the most likely reason Mark decided to apply for the new job? (1)

- A. Mark enjoys living in his current city.
B. Mark believes the new job is a better career opportunity.
C. The new job requires him to relocate to a different country.
D. Mark prefers jobs that offer lower salaries.
-

23. Meditation is a great way to enhance mental health and reduce stress, and you don't need a gym or special equipment to practice it. All you need is a quiet space, some time, and a comfortable place. Which of the following sentences best supports this idea? (1)

- A. Meditation is a simple, cost-effective practice that only needs a peaceful space to improve mental clarity.
B. Gym workouts are worse for stress relief than meditation because it needs a professional instructor.
C. Meditation requires special equipment like pillows or sound machines for effectiveness.
D. You don't need a quiet space, some time and comfortable position for meditation.
-

24. *Mary Jonson was an obstinate and isolated child, who spent her early years in the lush landscapes of India. Following the death of her neglectful parents, she was sent to live with her reclusive uncle, Mr. Craven, in an enormous, deserted mansion in England. Initially, Mary was disinterested in her surroundings, sulking in her loneliness. However, a fateful discovery of a hidden door in the garden wall sparked her curiosity. As Mary ventured into the long-forgotten garden, she encountered a wild, untamed world filled with neglected flora. Driven by a newfound purpose, she diligently nurtured the garden back to life, transforming it into a sanctuary of vibrant colors and charming scents. Through this process, she not only fostered the garden's revival but also cultivated friendships that helped her discover the joys of companionship, resilience, and personal growth.* (1)
How is Mary Jonson initially characterized in the passage?

- A. Stubborn and solitary B. Energetic and curious
C. Cheerful and sociable D. Compassionate and friendly
-

25. *same passage* (1)

What event led to Mary's relocation to her uncle's mansion?

- A. A family reunion B. Her desire for adventure
C. The death of her parents D. A travel to England for vacation

-
- 26. same passage** (1)
- What does Mary find within the hidden garden?
- A. A magical fountain
 - B. An overgrown, neglected space
 - C. A place filled with animated animals
 - D. A structured and organized landscape
-
- 27. same passage** (1)
- What transformation occurs in Mary as she cares for the garden?
- A. She gains a reputation as a gardener
 - B. She learns about gardening techniques
 - C. She becomes wealthier and more powerful
 - D. She discovers feelings of companionship and resilience
-
- 28. Every year, millions of monarch butterflies embark on an incredible migration from North America to central Mexico. This journey can span up to 3,000 miles, making it one of the longest migrations of any insect species. Monarchs travel in groups, known as clusters, and rely on specific environmental cues to guide them, such as temperature and the position of the sun. As they migrate, monarchs face numerous trials, including predators and changes in weather. They must find food sources, typically nectar from flowers, to sustain their journey. Interestingly, these butterflies are not born knowing the migration route; instead, it is believed they inherit the knowledge through instinct passed down from previous generations. Upon reaching their wintering grounds, the monarchs gather in vast numbers, covering the trees in a breathtaking display. This phenomenon not only underscores the resilience of the species but also highlights the importance of preserving their habitat. Protecting these migration pathways is crucial for the survival of monarchs, as environmental changes threaten their annual journey.** (1)
- What distinguishes the migration of monarch butterflies from those of other insect species?
- A. It happens only in small clusters.
 - B. It relies solely on the position of the sun.
 - C. It occurs without the need for food sources.
 - D. It spans up to 3,000 miles, making it one of the longest migrations.
-
- 29. same passage** (1)
- Which of the following best describes the challenges faced by monarch butterflies during migration?
- A. Lack of instinct on the migration route.
 - B. Sufficient food supply and safe climates.
 - C. Overpopulation at their wintering grounds.
 - D. Predators, adverse weather, and the search for nectar.
-
- 30. same passage** (1)
- In what way is the knowledge of the migration route significant for the monarch butterflies?
- A. It allows them to adapt to new climates.
 - B. It is vital for their survival during migration.
 - C. It enables them to find better feeding grounds.
 - D. It guarantees the success of their breeding cycle.

31. same passage

(1)

Based on the passage, what can be inferred about the role of environmental changes on the migration of monarch butterflies?

- A. Environmental changes have no significant impact on the migration of monarch butterflies.
- B. Monarch butterflies are capable of adapting their migration routes quickly in response to environmental changes.
- C. The survival of monarch butterflies during their migration is at risk due to environmental changes, making habitat preservation essential.
- D. Monarch butterflies rely solely on instinct for their migration and do not require external environmental cues.

32. The product of two numbers is 120 and the sum of their squares is 289. The sum of the numbers is?

(1)

- A. 20
B. 23
C. 169
D. 529

33. What is the value of X and Y respectively in a sequence 1, 5, 14, 30, 55, X, 140, 204, – – ?

(1)

- A. 85
B. 86
C. 91
D. 75

34. A library has an average of 510 visitors on Sundays and 240 on other days. The average number of visitors per day in a month of 30 days beginning with a Sunday is:

(1)

- A. 250
B. 276
C. 280
D. 285

35. Which of the following linear inequalities define the domain of the function $f(x,y)=\sqrt{4x-y+6}+\ln(2x+y-5)$?

(1)

- A. $4x - y \geq -6$ and $2x + y > 5$
- B. $4x - y \leq -6$ and $2x + y \geq 5$
- C. $4x - y \geq -6$ and $2x + y \geq 5$
- D. $4x - y \leq -6$ and $2x + y > 5$

36. Find an odd out from the given: 396, 427, 671, 264

(1)

- [illegible]

37. The White, Red and Black balls in a packet are in the ratio of 5:3:2. How many Red and Black balls are there, if there are 40 White balls?

(1)

- A. 30 reds and 50 black balls B. 16 reds and 24 black balls
C. 24 reds and 16 black balls D. 25 reds and 15 black balls

38. If $2a + b = x$ and $-2a + b = y$, then which one of the following expression is equal to ab ?

(1)

- A. $(x+y)/2$
B. $(x-y)/4$
C. $(x^2 - y^2)/8$
D. $(x^2 + y^2)/8$

39. The average monthly income of P and Q is Birr 5050. The average monthly income of Q and R is 6250 and the average of P and R is 5200. The monthly income of P is:

(1)

- A. 3500
B. 4000
C. 4050
D. 5000

40. If an equilateral triangle and a square have the same perimeter, what is the ratio of the length of square to the length of the side of equilateral triangle?

(1)

- A. 3:4
B. 1:2
C. 4:3
D. 2:1

41. Three numbers are in the ratio of 3:4:5 and the least common multiple is 2400. Their GCF is: (1)

A. 40
B. 80
C. 120
D. 200

42. A, B and C jointly thought of engaging themselves in business venture. A would invest Birr 6500 for 6 months, B Birr 8400 for 5 months and C Birr 10,000 for 3 months. A wants to be the working member for which he was to receive 5% of the profit. The profit earned was Birr 7400. What is the share of B in the profit? (1)

A. Birr 1900
B. 2660
C. 2800
D. 2840

43. In how many different ways can the letters of the word 'CORPORATION' be arranged, so that vowels always come together? (1)

A. 810
B. 1440
C. 2880
D. 50400

44. What is the probability of a number selected at random from the first 100 prime numbers is even? (1)

A. 99/100
B. 49/100
C. 1/100
D. 1/50

45. A committee consists of 6 men and 4 women. If two members are selected at random to form a subcommittee. What is the probability that both members are women? (1)

A. 1/5
B. 2/15
C. 3/10
D. 1/10

46. If average age of husband, wife and their child 3 years ago was 27 years and that of wife and child 5 years ago was 20 years. The present age of the husband is: (1)

A. 35 years
B. 40 years
C. 50 years
D. 90

47. $(287)^2 + (269)^2 - 2(287 \times 269)$ is equal to: (1)

A. 534
B. 446
C. 354
D. 324

48. A piece of machinery is worth Birr 100,000 today and loses 25% of its value each year. How much will it be worth after 4 years? (1)

A. 75000
B. 41640.625
C. 31642
D. 31640.625

49. For x is a real number, compare Quantity A = $\sqrt{4x^2+4x+1}$ and Quantity B = $2x+1$. (1)

A. Quantity A is less than quantity B
B. Quantity A is greater than quantity B
C. Quantity A is equal to quantity B
D. The relationship cannot determine

50. Compare the two quantities: Quantity A = percent of income increase from 75 to 85 thousand birr; Quantity B = percent of income increase from 200 to 220 thousand birr. (1)

A. Quantity A is equal to quantity B
B. Quantity A is greater than quantity B
C. Quantity A is less than quantity B
D. The relationship cannot determine

51. The table below shows the sales of computers manufactured by a company over the years. Number of Different Types of Computer Sold by a Company over the Years (Numbers in Thousands) | Year | 4hp | 7hp | 32hp | 35hp | 55hp | Total | |-----|-----|-----|-----|-----|-----|-----| | 2000 | 75 | 144 | 114 | 102 | 108 | 543 | | 2001 | 90 | 126 | 102 | 84 | 126 | 528 | | 2002 | 96 | 114 | 75 | 105 | 135 | 525 | | 2003 | 105 | 90 | 150 | 90 | 75 | 510 | | 2004 | 90 | 75 | 135 | 75 | 90 | 465 | | 2005 | 105 | 60 | 165 | 45 | 120 | 495 | | 2006 | 115 | 85 | 160 | 100 | 145 | 605 |
- What was the approximate percentage increase in the sales of 55hp computer in 2006 compared to that in 2000?
- A. 28% B. 31%
C. 33.26% D. 34.26%
-
52. same table (1)
Which type of computers sales is maximum over seven years?
- A. 4hp B. 7hp
C. 32hp D. 35hp
-
53. same table (1)
What is the difference in the number of 35hp computers sold in 2001 and 2005?
- A. 24000 B. 28000
C. 35000 D. 39000
-
54. same table (1)
Which types of computer was a continuous decrease in sales from 2000 to 2005?
- A. 4hp B. 7hp
C. 32hp D. 35hp
-
55. same table (1)
The percentage of 4hp computer sold to the total number of computer sold was maximum in the year?
- A. 2002 B. 2004
C. 2003 D. 2005

— END OF EXAMINATION —